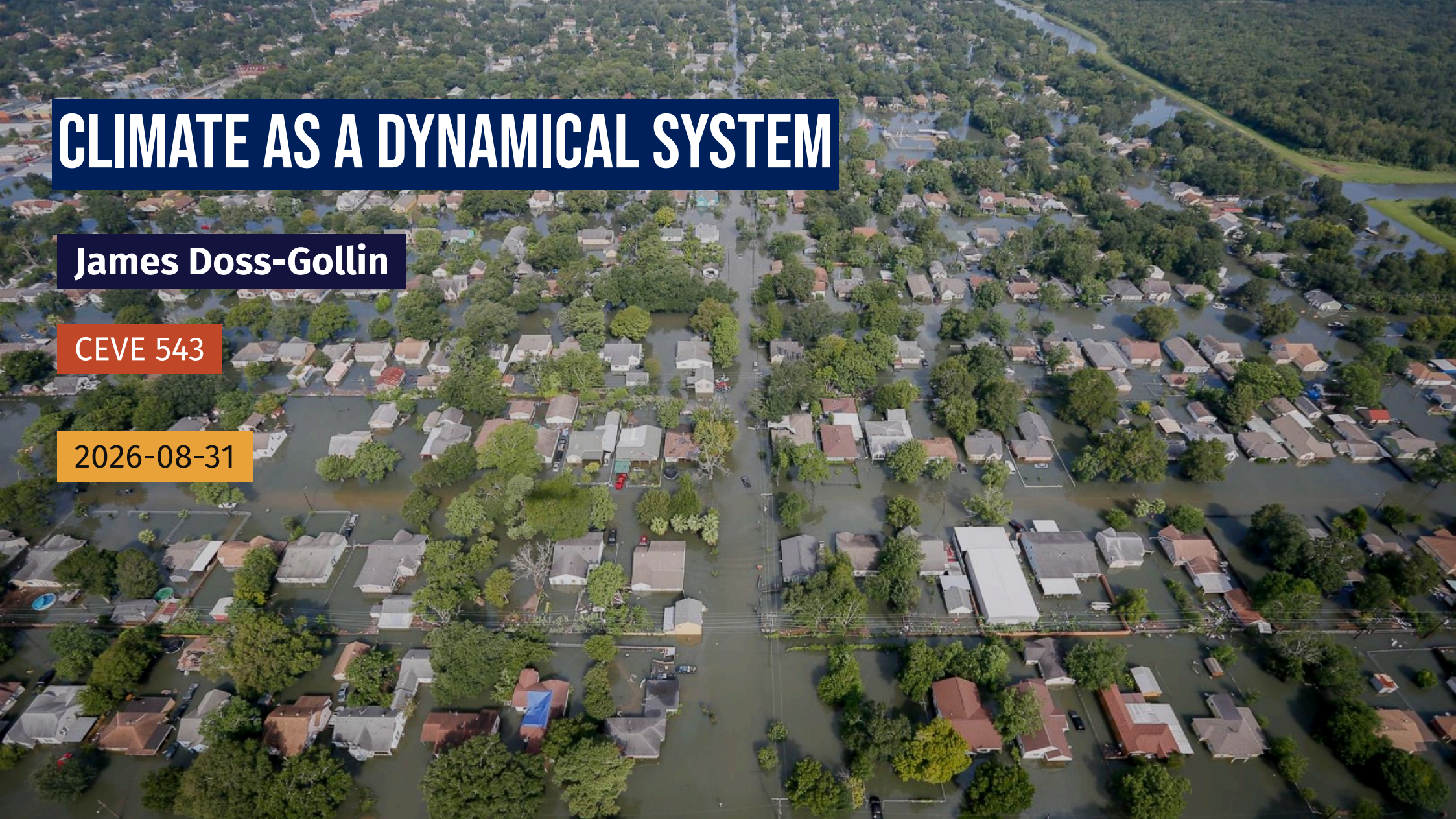


CLIMATE AS A DYNAMICAL SYSTEM

An aerial photograph of a suburban neighborhood completely inundated with floodwater. The water is a murky, brownish-grey color, covering the streets, lawns, and yards. Numerous houses with various roof colors (brown, grey, red) are visible, some with water reaching their windows. Trees are scattered throughout the landscape, some partially submerged. In the upper right corner, a green golf course is visible, partially surrounded by water. The overall scene depicts the severe impact of flooding on a community.

James Doss-Gollin

CEVE 543

2026-08-31

WHY SIMPLE MODELS?

Deterministic dynamical systems, with and without random forcing, can produce much of the behavior we saw in the first deck: persistence, clustering, apparent trends, regime-like shifts.

These models help you build intuition about what kinds of behavior are possible and how to describe them mathematically.

A DYNAMICAL SYSTEM

A dynamical system has a **state** $X = (X_1, \dots, X_p)$ and a **rule** F that determines how the state changes. In continuous time:

$$\dot{X}_i = F_i(X_1, \dots, X_p), \quad 1 \leq i \leq p$$

In discrete time:

$$X_{t+1} = G(X_t)$$

The system is **nonlinear** when the rule involves products or powers of the state variables, like $x \cdot z$ or $x(1 - x)$. Linear systems cannot produce chaos.

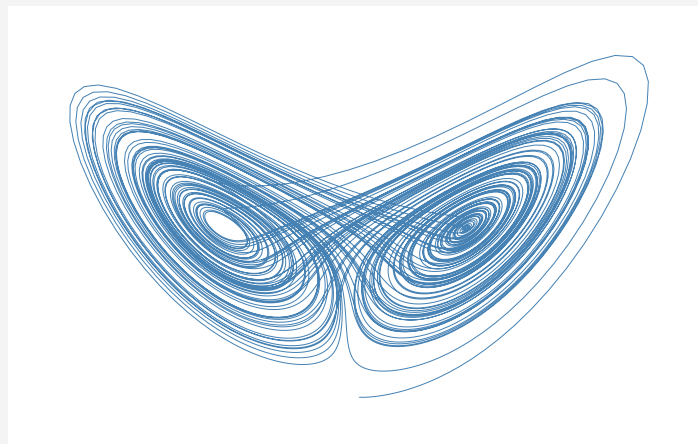
IMPORTANT TERMS

- **Trajectory:** the sequence of states the system visits over time.
- **Phase space:** the space of all possible states.
- **Attractor:** the set of states the system settles onto over long times.
- **Sensitive dependence:** small changes in initial conditions lead to diverging trajectories.

LORENZ AND SENSITIVE DEPENDENCE

In 1961, Edward Lorenz restarted a weather simulation from a printout that rounded the state to three decimal places instead of the six stored in memory. The restarted run diverged completely from the original within a few simulated days.

For supplementary notes on this discovery, read James Gleick's *Chaos* (1987), chapters 1-2.



$$\dot{x} = 10(y - x)$$

$$\dot{y} = x(28 - z) - y$$

$$\dot{z} = xy - \frac{8}{3}z$$

THE LOGISTIC MAP

A model of bounded growth:

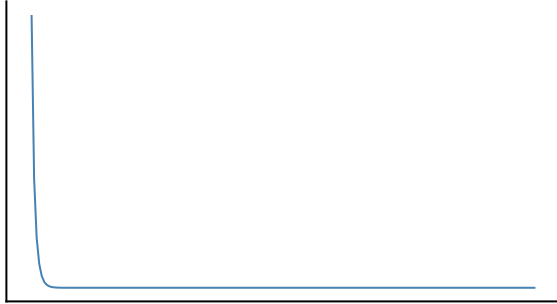
$$x_{t+1} = r x_t (1 - x_t)$$

The parameter r controls how strongly the system overshoots.

```
function logistic(r; n=200, x0=0.4)
    x = zeros(n)
    x[1] = x0
    for t in 1:(n - 1)
        x[t + 1] = r * x[t] * (1 - x[t])
    end
    return x
end
```

SIX VALUES OF r , SIX BEHAVIORS

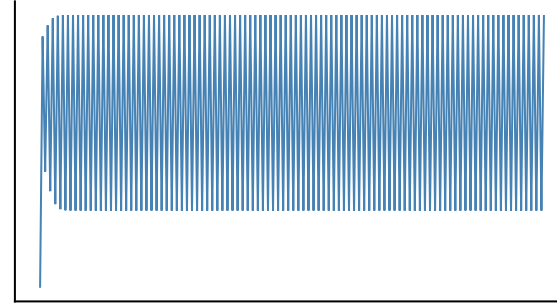
$r = 1.5$ (fixed point)



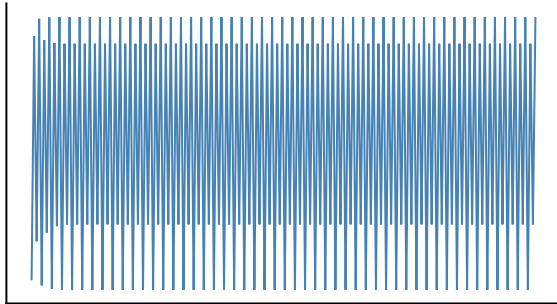
$r = 2.8$ (fixed point)



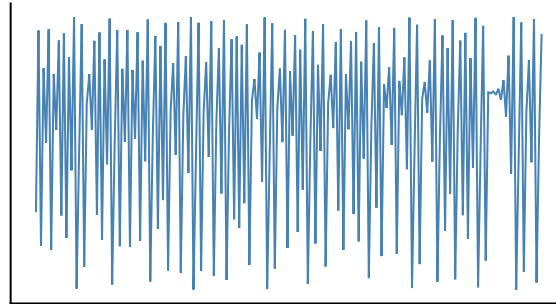
$r = 3.2$ (period 2)



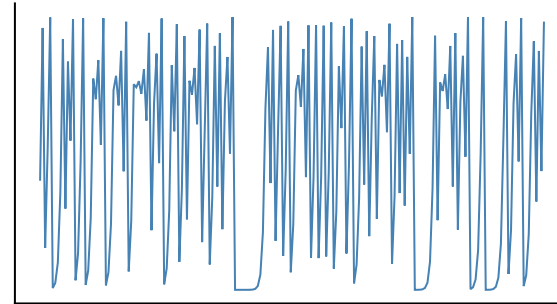
$r = 3.5$ (period 4)



$r = 3.8$ (chaos)



$r = 4.0$ (chaos)



WHAT DO WE SEE?

A **fixed point**: the system settles to a constant.

A **periodic orbit**: the system repeats exactly every k steps.

Chaos: the system is deterministic but sensitive to initial conditions, so small differences grow exponentially and the trajectory never repeats.

MEASURING MEMORY

The **autocorrelation** at lag h measures how much today's value tells you about the value h steps later (Kantz, 2004, §2.2):

$$\rho_h = \frac{\text{Cov}(X_t, X_{t+h})}{\text{Var}(X_t)}$$

If $\rho_h = 0$, the process has no memory at lag h . If $\rho_h > 0$, high values tend to follow high values.

WHAT ABOUT RANDOMNESS?

Processes no model can predict force real systems: weather, wind bursts, volcanic eruptions.

Adding a noise term ε_t to a deterministic rule gives a **stochastic dynamical system**.

DISCUSSION

Given a single observed record of finite length: why is it hard to reconstruct the system that produced it?

LOGISTICS

- Lab 1 due Friday
- Set up the software for Lab 2 before Friday's session
- I posted this week's practice problems